The purpose of this section is to prove the following:

Theorem 0.1. *Let X be a smooth variety of general type and $a : X \longrightarrow A$ be its Albanese morphism. If A is simple, then any holomorphic 1-form $w \in H^0(X, \Omega^1_X)$ has non-empty zero set.*

We will need several preliminary results:

Proposition 0.2. *Let X be a projective variety and $a : X \longrightarrow A$ be a morphism to an abelian variety. If $Z := a(X)$ is a variety of general type, then every holomorphic 1-form $w \in a^*H^0(A, \Omega^1_A) \subset H^0(X, \Omega^1_X)$ has non-empty zero set.*

Proof. It suffices to show that any holomorphic one form $w' \in H^0(A, \Omega^1_A)$ restricted to Z has non-empty zero set. Let $W \subset H^0(A, \Omega^1_A)$ be the set of those holomorphic 1-forms $w' \in H^0(A, \Omega^1_A)$ such that $w'|Z$ vanishes at some point $z \in Z$. It is easy to see that W is closed (in the euclidean topology) and so it suffices to show that W is dense in $H^0(A, \Omega^1_A)$.

If for a general (smooth) point $z \in Z$, one has $T_z(Z)$ is contained in a fixed hyperplane of $T_z(A)$ then, near z, one has that Z is contained in a subgroup of A and hence Z is globally contained in a subgroup of A. This is impossible as Z generates A. □

Proposition 0.3. *Let X be a projective variety and $a : X \longrightarrow A$ be a morphism to an abelian variety. If a is surjective and not smooth over an irreducible divisor D of general type, then every holomorphic 1-form $w \in a^*H^0(A, \Omega^1_A) \subset H^0(X, \Omega^1_X)$ has non-empty zero set.*

Proof. Consider $W \subset H^0(A, \Omega^1_A)$ the set of those holomorphic 1-forms $w \in H^0(A, \Omega^1_A)$ such that $a^*w \in H^0(X, \Omega^1_X)$ vanishes at some point $x \in X$. As above, W is closed in the Euclidean topology and so it suffices to show that it is dense.

Consider $D^0 \subset D$ a (non-empty) open set such that for all $z \in D^0$ $rank(da_x) = g-1$ for some $x \in X_z$ (cf. [Ha] III Proposition 10.6) and D is smooth at z. Let $x_1, ..., x_g$ be local coordinates of A at z such that D is defined by $x_g = 0$ and $w = w_z \in H^0(A, \Omega^1_A)$ such that $w(z) = dx_g$. Then $w|D$ vanishes at z and a^*w vanishes at some point $x \in X$ such that $a(x) = z$ (in fact at any such point with $rank(da_x) = g-1$). Since D is of general type, (reasoning as in the previous proposition) one easily sees that the set $\{w_z | z \in D^0\} \subset W$ is dense in $H^0(A, \Omega^1_A)$. □

Proposition 0.4. *Let $f : X \longrightarrow A$ be an algebraic fiber space, A an abelian variety and m a positive integer such that such that $f_*(\omega^{\otimes m}_{X/A})$ is big, then f is not smooth in codimension* 1.

Proof. Since $f_*(\omega^{\otimes m}_{X/A})$ is big, there exists an ample line bundle H on A and an integer $a > 0$ such that $\hat{S}^a(f_*(\omega^{\otimes m}_{X/A})) \otimes H^\vee$ is big. Let $m_k : A_k = A \longrightarrow A$ be multiplication by an integer k, so m_k is an étale map such that $m_k^*H = (H')^k$ with H' an ample line bundle on A_k. Let $g = \dim(A)$, $k = ma3r(g-1)$ and $r = \dim(X) - \dim(A)$. Let

$$f' : X' := X \times_A A_k \longrightarrow A_k = A.$$

Then

$$m_k^*(\hat{S}^a(f_*(\omega^{\otimes m}_{X/A})) \otimes H^\vee) = \hat{S}^a({f'}_*(\omega^{\otimes m}_{X'/A})) \otimes ({H'}^{3r(g-1)})^{-ma}$$

is big (cf. [Mo] (5.1.1) (d)) and hence ${f'}_*(\omega^{\otimes m}_{X'/A}) \otimes ({H'}^{3r(g-1)})^{-m}$ is also big. Since H' is ample, $3H'$ is very ample. Let C be the curve obtained by intersecting $g-1$ general elements in $|3H'|$ and $\mathcal{A} := (3r(g-1)H')|C$. Then

$$\deg(\omega_C) = (g-1)(3H')^g \quad and \quad \deg(\mathcal{A}) = r(g-1)(3H')^g = r\deg(\omega_C).$$

Let $Y=(f')^{-1}(C)$ and $h: Y \longrightarrow C$. If f is smooth in codimension 1, then f' is smooth in codimension 1 and so h is smooth. Since

$$\left(f'_*(\omega_{X'/A}^{\otimes m}) \otimes (H'^{3r(g-1)})^{-m}\right)|C = h_*(\omega_{Y/C}^{\otimes m}) \otimes \mathcal{A}^{-m},$$

it follows that $h_*(\omega_{Y/C}^{\otimes m}) \otimes \mathcal{A}^{-m}$ is also big and hence ample. By [VZ] Proposition 4.1 (with $\delta=0$, $s=0$), one has

$$\deg(\mathcal{A}) < \dim(Y/C)\deg(\omega_C) = r\deg(\omega_C).$$

This is impossible and hence f is not smooth in codimension 1. □

Proof. (of Theorem 0.1) Since A is simple, by a result of Ueno (cf. [Mo] Theorem 3.7) any proper subvariety of A is of general type. By Proposition 0.2, if there is a $w \in H^0(X, \Omega_X^1)$ with empty zero set, then $a: X \longrightarrow A$ is surjective and by Proposition 0.3, a is smooth in codimension 1. Let $X \longrightarrow Z \longrightarrow A$ be the Stein factorization, then $Z \longrightarrow A$ is smooth and hence étale in codimension 1 and so Z is birational to an abelian variety and hence Z is birational to A. It follows that $a: X \longrightarrow A$ is an algebraic fiber space. Since X is of general type, $a_*(\omega_{X/A}^m) = a_*(\omega_X^m)$ is big for some $m>0$, but by Proposition 0.4 this is impossible. □

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